

# Quadrilaterals & Other Polygons

US Common Core – High School Geometry

Domain: Congruence (G-CO.C.11) | Expressing Geometric Properties with Equations (G-GPE.B)

## 1 Polygon Basics

### Definitions

A **polygon** is a closed figure with many straight sides. A **diagonal** of a polygon is a line segment inside the polygon connecting two non-consecutive vertices.

### Theorem 1 (Polygon Interior Angles Theorem)

The sum of the interior angles of a polygon with  $n$  sides is

$$(n - 2) \times 180^\circ.$$

### Example

For a triangle,  $n = 3$ :  $(3 - 2) \times 180^\circ = 180^\circ$ .

### Theorem 2 (Polygon Exterior Angles Theorem)

The sum of the exterior angles of a polygon (one at each vertex) is always  $360^\circ$ .

### Example

A quadrilateral has exterior angles  $x^\circ$ ,  $108^\circ$ ,  $121^\circ$ ,  $59^\circ$ :

$$x + 108 + 121 + 59 = 360 \Rightarrow x = 72^\circ.$$

## 2 Types of Polygons

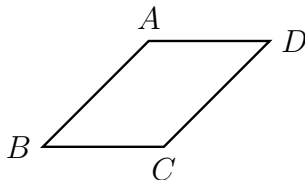
### Definitions

- **Equilateral polygon**: all sides congruent with each other.
- **Equiangular polygon**: all angles congruent with each other.
- **Regular polygon**: both equilateral *and* equiangular.

### 3 Parallelograms

#### Definition

A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel to each other.



#### Theorem 3 (Opposite Sides Theorem)

If a quadrilateral is a parallelogram, then the opposite sides of the quadrilateral are congruent with each other.

#### Theorem 4 (Opposite Angles Theorem)

If a quadrilateral is a parallelogram, then the opposite angles of the quadrilateral are congruent with each other.

#### Example

Parallelogram  $ABCD$ : side  $AD = x + 4$  opposite side  $BC = 12$ ;  $\angle A = y$ , and the angle at  $C$  adjacent to  $BC$  is  $65^\circ$  (so  $\angle A$  and this angle are alternate, giving  $y = 65^\circ$ ):

$$x + 4 = 12 \Rightarrow x = 8, \quad y = 65^\circ.$$

#### Theorem 5 (Converse of Opposite Sides)

If the opposite sides of a quadrilateral are congruent with each other, then the quadrilateral is a parallelogram.

#### Theorem 6 (Converse of Opposite Sides Parallel)

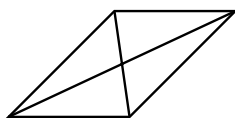
If the opposite sides of a quadrilateral are parallel with each other, then the quadrilateral is a parallelogram.

#### Theorem 7 (Converse of Opposite Angles)

If the opposite angles of a quadrilateral are congruent with each other, then the quadrilateral is a parallelogram.

### Theorem 8 (Bisecting Diagonals Theorem)

If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.



## 4 Rectangles, Rhombi, and Squares

### Definitions

- A **rectangle** is a parallelogram with all four angles right angles.
- A **rhombus** is a parallelogram with all sides congruent.
- A **square** is both a rhombus *and* a rectangle.

### Rectangle Corollary

A quadrilateral is a rectangle if and only if its four angles are right angles.

### Rhombus Corollary

A quadrilateral is a rhombus if and only if the four sides are congruent with each other.

### Square Corollary

A quadrilateral is a square if and only if it is both a rhombus and a rectangle.

### Diagonals of a Rectangle Theorem

If a quadrilateral is a rectangle, then its diagonals bisect each other.

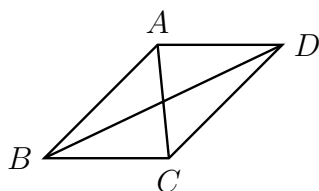
*Note: A quadrilateral is a rectangle if and only if its diagonals are congruent with each other.*

### Diagonals of a Rhombus Theorem

The diagonals of a rhombus are perpendicular with each other.

### Angles of a Rhombus Theorem

The angles of a rhombus are bisected by their diagonals.

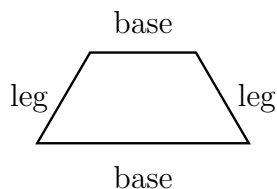


## 5 Trapezoids and Kites

### Trapezoid Vocabulary

A **trapezoid** has exactly one pair of parallel sides, called the **bases**; the two non-parallel sides are the **legs**.

- **Isosceles trapezoid**: the legs are congruent.
- **Regular (scalene) trapezoid**: the sides are all of unequal length.



### Theorem 1 (Isosceles Trapezoid Base Angles Theorem)

Each pair of base angles of an isosceles trapezoid are congruent with each other.

### Theorem 2 (Converse)

If only one pair of the base angles of a trapezoid is congruent with each other, then the trapezoid is an isosceles trapezoid.

### Theorem 3

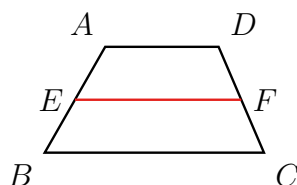
A trapezoid is an isosceles trapezoid if and only if its diagonals are congruent with each other.

### Theorem 4 (Trapezoid Midsegment Theorem)

The midsegment of a trapezoid is parallel to its bases, and its length is half the sum of the lengths of the bases:

$$EF = \frac{1}{2}(AD + BC)$$

where  $E, F$  are the midpoints of the legs and  $AD, BC$  are the bases.



### Kite

A **kite** is a quadrilateral with two pairs of consecutive congruent sides (adjacent sides equal, rather than opposite sides).

#### Theorem 5 (Diagonals of a Kite Theorem)

The diagonals of a kite are perpendicular with each other.

#### Theorem 6 (Angles of a Kite Theorem)

If a quadrilateral is a kite, then exactly one pair of its opposite angles are congruent with each other.

